

Exploratory Data Analysis of IPL Ball-by-Ball Data

Minor Project Report | Data Science & Business Analytics Traineeship Program

Project Objective

The objective of this project is to perform an in-depth Exploratory Data Analysis (EDA) on IPL ball-by-ball delivery data. The project focuses on cleaning and preprocessing the dataset, exploring batting, bowling and team-level patterns, identifying trends and relationships, and presenting meaningful findings through visualizations.

Dataset Description

The project uses **deliveries.csv**, an IPL ball-by-ball dataset where each row represents a delivery. It includes match ID, innings, teams, over, ball, batsman, bowler, runs, extras, dismissal information and fielder details. The original dataset contains **150,460 rows and 21 columns**.

Tools & Libraries Used

Python and **Jupyter Notebook** are used for the workflow. **Pandas** and **NumPy** support data handling and preprocessing, while **Matplotlib** and **Seaborn** are used for visual analysis.

Project Workflow

1	Data Import	Load deliveries.csv using Pandas and preview records with df.head() .
2	Data Understanding	Inspect shape, columns, data types, descriptive statistics, unique values and general dataset information.
3	Missing Values	Check null values. Dismissal-related nulls are expected because most deliveries do not produce a wicket, so they are handled meaningfully instead of dropping valid deliveries.
4	Duplicates	Identify duplicate rows, remove confirmed duplicates and verify that the duplicate count becomes zero.
5	Preprocessing	Review data types and potential outliers. Legitimate cricket events such as boundaries are retained even when statistically unusual.
6	EDA & Visualization	Create 10+ meaningful visualizations covering batsmen, bowlers, teams, scoring, wickets, extras, trends and correlations.
7	Export	Save the cleaned DataFrame as cleaned_deliveries.csv and retain the completed notebook for submission.

EDA Visualizations

The analysis includes more than ten visualizations: **top batsmen by runs, top bowlers by wickets, team-wise runs, runs-per-delivery distribution, fours vs sixes, top six-hitters, dismissal types, over-wise scoring trend, extras analysis, correlation heatmap, a boxplot and top fielders by catches**.

Key Analysis Approach

Bowling wickets are calculated only from dismissal types credited to the bowler. Missing dismissal fields are interpreted in their cricket context. Statistical outliers are reviewed before removal because values such as four and six runs are valid cricket events.

Challenges Faced & Solutions

File path: the initial `FileNotFoundError` was solved by using the correct Windows path. **Missing values:** dismissal-related nulls are handled without deleting valid deliveries. **Outliers:** unusual but legitimate cricket observations are retained after contextual inspection.

Conclusion

The project follows a complete EDA workflow from data import through understanding, cleaning, preprocessing, visualization, interpretation and export. The final submission consists of the **cleaned CSV dataset**, the **complete Jupyter Notebook** and the **one-page project report**.